

Effect of Auditory Environment on Mental Workload & Task Performance: non-invasive mobile neuroimaging enables this study in naturalistic environments

Functional Hemispheric Asymmetry and Multisensory Neurofeedback Training to Improve Brain Health

A Scoping Review Protocol and Research Framework

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Abstract

[TODO: Write structured abstract last (Background / Objective / Methods / Anticipated results). Keep to ~250 words; follow the PRISMA-ScR abstract checklist.]

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1 Introduction and Rationale

1.1 Problem statement

- The brain is constantly processing the entire auditory environment, yet attention and task performance is still achieved.
- We know that noisy environments are distracting and reduces performance, but how and why?
- Gold-standard fMRI is itself a noisy machine requiring supine subjects, thus mobile neuroimaging enables more naturalistic study of these effects.
- EEG-fNIRS can enable this study.

2 Literature notes

[TODO: One block per PDF in literature/. Fill in the *Notes* paragraph as you read; add \items for key findings, methods, sample, relevance.]

2.1 moore2017fatigue

moore2017fatigue (moore2017fatigue) — moore2017fatigue (moore2017fatigue)
literature/moore2017fatigue.pdf

- EEG/ERP (ERN, N1) + self-report + RT during 50-min auditory choice task; fatigue → reduced N1 (arousal) and decreased activation in dorsal attention network. Normal-hearing adults (n=19).

Notes [TODO: add notes]

2.2 alho2014auditorycortex

alho2014auditorycortex (alho2014auditorycortex) — alho2014auditorycortex (alho2014auditorycortex)
literature/alho2014auditorycortex.pdf

- Meta-analysis of fMRI studies: stimulus-dependent vs. attention-related modulation of auditory cortex. Useful background on where/how attention shapes auditory processing.

Notes [TODO: add notes]

2.3 yun2025anc

yun2025anc (yun2025anc) — yun2025anc (yun2025anc)
literature/yun2025anc.pdf

- fNIRS; active noise cancellation vs. noise during concentration tasks; neurophysiological effect of ANC in noisy environments. Very close to the target question.
- Auditory noisy environments is clearly linked to decreased task performance in humans, likely a consequence of increased stress and fatigue, yet neurophysiological evidence for the mechanisms are not yet unassailable.

- Under noisy conditions the PFC has been shown to increase activity, potentially leading to excessive and unproductive mental workload.
- Inclusion criteria: pure-tone average (PTA) of less than 25 dB at 500, 100, 2000 and 4000 Hz.
- Objective task performance + subjective questionnaire
- HRFs via block-averaged fNIRS data -3 to +55 seconds to task onset
- Brain activation quantified by β via GLM with a modeled double gamma HRF and
- ANC ON vs ANC OFF in a two sample t-test. (+ point biserial correlation coefficient to gauge the time axis).
- channel significance a Wilcoxon signed-rank test. Benjamin-Hochberg FDR correction, $\alpha = 0.05$
- ANC ON produced significantly higher task performance than ANC OFF, but not significant differences in behavioural performance such as reaction time. Indicating the cognitive load.
- 3 of 4 channels with statistically significant difference between the two conditions were located in the right hemisphere (which corresponds to the literature about the right DLPFC in cognitive task and sensitivity to cognitive load (MW)).
- Specifically, this study observed strong deactivation of HbO during the task, I believe this should nuance a little. If the ANC ON had LOWER HbO than ANC OFF, what does that mean?
- Subjective effort was significantly lower for ANC ON.
- They conclude ANC OFF lead to unnecessary consumption of neural resources, in light of the reduced task performance, this indicates ANC as effective to alleviate excessive cognitive load during auditory noise.
- However, this was a relatively simple task without complexity, we know from the literature that neural efficiency and MW is sensitivity to task difficulty.
- Their interpretation, that the stronger PFC deactivation during ANC OFF means that more of the brain resources was dedicated to the actual task, rather than separating speech from noise, thus reflecting the results. This opens a very interesting addition, temporal and parietal channels.
- Further indicates right PFC as a sensitive area to cognitive loads and MW.
- Recommends physiological measurements to avoid the variability in subjective VAS listening effort measures.

Notes [\[TODO: add notes\]](#)

2.4 han2023entrainment

han2023entrainment (han2023entrainment) — han2023entrainment (han2023entrainment)
 literature/han2023entrainment.pdf

- Combined acoustic + visual (acousto-optic) 40-Hz stimulation to drive coherent gamma entrainment. Relevant to the neurofeedback / multisensory-stimulation arm.

Notes [TODO: add notes]

2.5 hiwa2018fnirs

hiwa2018fnirs (hiwa2018fnirs) — hiwa2018fnirs (hiwa2018fnirs)

literature/hiwa2018fnirs.pdf

- fNIRS; auditory environments (e.g. silence/noise/music) vs. intellectual work performance; neural correlates in PFC.
- How music affects performance varies according to both the music and individual.
- The efficacy of intellectual work is closely related to brain activation.
- Musical effect on intellectual work has been reported, but naturalistic noise environments are yet underreported.
- Time of day + room humidity and temperature reported, good.
-

Notes [TODO: add notes]

2.6 gabbard2017fnirs

gabbard2017fnirs (gabbard2017fnirs) — gabbard2017fnirs (gabbard2017fnirs)

literature/gabbard2017fnirs.pdf

- fNIRS (PFC HbO) + eye tracking + RT/accuracy in simulated noisy ISR work; short-intermittent noise → larger PFC activation than long-intermittent noise.
- Auditory noise increases cognitive load, potentially reducing work performance by affecting attention, working memory, etc.
- Again highlights that this doesn't necessarily hold for very simple tasks, touching neural efficiency theory.
- MW can be assessed by PFC-based fNIRS. Stress can reduce PFC activation.
- Again $\alpha = 0.05$
- Their findings are not clear regarding the "Noise leads to increased PFC activity" which I questioned immediately and noted a question mark in my notes.
- Again task complexity is nuanced. It seems intermittent noise or bursts are more detrimental than sustained noise.
- fNIRS is deemed a major candidate for military applications to assess MW.
- Specifically this study found different types of noise affect the brain activation, supported by the Fan et al?
- Rest-Memorize-Rest/Retain-Answer-Rest design, 8 digit short-term memory recall task. Again uses subjective VAS for pleasantness, objective physiology would be beneficial here.
- Used Hitachi fNIRS + POTATO for tomography analysis. Brain activation quantized via z-score. Functional connectivity by Pearson's into Fisher's Z transform.

- Again alpha 0.05.
- Subjective pleasantness was significantly higher in silence vs white noise.
- Motor and visual cortical areas showed significantly stronger connectivity in silence vs noise.
- Strong individual differences led the analysis to consider 3 cases: positive responders, neutral and negative responders to the noise.
- For the positive responders (White group) no significant brain activation difference between noise and silence was found. For the average and silence groups the brain activation differed significantly between noise and silence in the right superior temporal gyrus, subcentral area, frontal pole, dorsolateral prefrontal cortex, and inferior frontal gyros. Again DLPFC seems to be a sensitive area to MW.
- The silence group showed a significant difference in both temporal lobe with increased activation in silence.
- The average group showed small difference in the pre-motor, SMA areas between noise and silence
- The white group showed significantly increased connectivity in the temporal areas during noise. The white group showed high interhemispheric connectivity during noise vs silence.
- Average and Silence group showed significantly stronger PMA-SMA connectivity in silence versus noise (IMPORTANT), indicating motoric areas are downregulated during noise.
- The silence group PMA-SMA autoconnectivity increased in noise, and
- Silence group showed higher DLPFC and PFC connectivity during silence vs noise (BIG)
- The White group shows complex habituation and compensation strategies enabling higher performance.
- The brain activation between environments differed significantly of the Silence group, not for the white group.
- The average and silence groups showed decreased PFC-PMA-SMA connectivity during white noise versus silence (BIG). Here we see individual differences in neural efficiency or compensatory behaviour, the average group's performance did not suffer from noise where the silence group did perform poorer during noise. Overall, the effect of noise can be individual and fNIRS may be an appropriate tool to gauge how an individual responds to noise.

Notes [\[TODO: add notes\]](#)

2.7 ravi2021fnirs

ravi2021fnirs (ravi2021fnirs) — ravi2021fnirs (ravi2021fnirs)
 literature/ravi2021fnirs.pdf

- Review-style paper on using fNIRS for cognitive workload assessment; surveys subjective vs. neuroimaging workload measures.

Notes [\[TODO: add notes\]](#)

2.8 pieper2021noise

pieper2021noise (pieper2021noise) — pieper2021noise (pieper2021noise)
literature/pieper2021noise.pdf

- EEG + subjective workload under environmental noise vs. noise-cancellation while working; EEG-based workload assessment.

Notes [\[TODO: add notes\]](#)

2.9 fan2022noise

fan2022noise (fan2022noise) — fan2022noise (fan2022noise)
literature/fan2022noise.pdf

- Noise exposure $\times$ mental workload; multiple physiological responses during task execution (check which modalities—EEG/ECG/EDA?).
- Acoustic noise defined is disturbing, unpleasant or unwanted sound.
- Acoustic noise exposure is associated with negative health effects
- Physiology reacts significantly to noise above 70 db.
- 35 to 65 db induces annoyances and subsequently can alter sleep patterns.
- 66 to 85 db can cause significant impairment including psychological, neurological and hearing damage.
- 85 to 115 db can cause significant hearing damage and psychological disturbance.
- Mental workload (MW) in an optimal range improves human performance, where auditory noise is a significant factor in MW. The mind wants an optimal range of MW, not under or overloaded. Flow state means an optimal amount of challenge to promote fun, learning and performance.
- MW can be measured in many objective and subjective methods, the article points to ECG (HRV), EEG and eye movements.
- Many have focused on dB meters and their effects, but rather we are interested in the characteristics of the acoustic noise and what components constitute the damaging or promotion of MW.
- A cognitive task experiment was conducted with 3 noise levels N85-S1, N80-S1, and N75-S2, where N indicates decibel and S represents sharpness level.
- A NASA-authored cognitive task at different difficulties were given.
- ECG, EEG and eye tracking was monitored. I will only confer ECG and EEG results.
- Overall, the subjects used additional effort to complete the tasks under the noisy conditions and noise level contributes to MW.
- Noise intensity significantly increased HRV.
- Increased MW level decreased alpha power and increased theta power.
- Increased stress significantly increased HRV and theta bandpower. -

- Sharper noise decreased alpha power, i.e increase stress.
- As MW increases the body compoesnts by increasing effort and stress, upto a given point where stress eentually impairs performance.
- Takeaway: Use ECG, EEG, fNIRS. Prioretize HRV, alpha & theta band power, and PFC.

Notes [\[TODO: add notes\]](#)

2.10 cause2017workload

cause2017workload (cause2017workload) — **cause2017workload (cause2017workload)**
 literature/cause2017workload.pdf

- fNIRS in PFC; mental workload and neural efficiency (piloting-type tasks); good methodological reference for fNIRS workload quantification.
- fNIRS is suitbale to measure MW and neural efficeancy in both labratroy and ecological settings.
- As the fNIRS signal did correlate strongly with percieved MW, but not actual task-perforamcen, indicates the already established existance of neural effieancy.
- The neural efficacy index may be valid across indivudulas engaging in the same cognitive task, but breaks down when different cognitive domains are at play.
- Increased task difficulty correletes to increased HbO and decreased HbR, indicating fNIRS are relevant for mental effort measurement.
- However, predicting task performance from fNIRS alone is not feasable due to varaiblity in neural efficacy.
- See also (**neubauer2009neuralefficiency**) (neural efficiency review) and §3.

Notes [\[TODO: add notes\]](#)

2.11 rosenkranz2024eeg

rosenkranz2024eeg (rosenkranz2024eeg) — **rosenkranz2024eeg (rosenkranz2024eeg)**
 literature/rosenkranz2024eeg.pdf

- Mobile EEG (ERPs, TRFs) during simulated laparoscopic surgery with OR soundscape; auditory work strain under low/high cognitive demand (n=22). Strong naturalistic-mobile-neuroimaging exemplar.

Notes [\[TODO: add notes\]](#)

2.12 noboa2025entrainment

noboa2025entrainment (noboa2025entrainment) — **noboa2025entrainment (noboa2025entrainment)**
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- Neural entrainment to beat + working memory predict sensorimotor synchronisation; relevant to auditory entrainment mechanisms.

Notes [TODO: add notes]

2.13 cannon2026surgery

cannon2026surgery (cannon2026surgery) — cannon2026surgery (cannon2026surgery)
literature/cannon2026surgery.pdf

- fNIRS + Surg-TLX + modified System Usability Survey used to quantify cognitive load across three percutaneous posterior spine surgical workflows; $n = 6$ surgeons.
- Excessive cognitive load may negatively affect surgical performance.
- Subjective cognitive-load assessments are valuable but retrospective and prone to influence by other factors; a multimodal approach is therefore needed for real-time cognitive-load assessment.
- HRV is the most common objective measure, but fNIRS is establishing itself as a complementary one.
- Aim is not only to quantify and reduce mental fatigue and stress, but also to guide development and integration of technology into the operating room, coupling task performance with objective and subjective measures.
- Compared robot-assisted (ARA), freehand (FHN) and fluoroscopy-guided (FG) workflows.
- Assumes greater PFC oxygenation means greater cognitive load — note the nuance here: this depends on task complexity and the individual (see (neubauer2009neuralefficiency; cause2017workload)).
- fNIRS metrics: total peak oxygenation, duration-dependent oxygenation, average oxygenation per second (FG vs. ARA) and total peak haemoglobin.
- Null hypothesis: mean of FHN equals ARA.
- Result: the fNIRS metrics are significant predictors of surgical workflow and the null hypothesis is rejected; ARA significantly reduced cognitive load.
- Takeaway: workflow/technology choice measurably shifts PFC load — a precedent for treating an environmental factor (here the workflow, for us the auditory environment) as the manipulated variable in a naturalistic fNIRS design. Pairs with (rosenkranz2024eeg) (same OR setting, EEG, soundscape as the manipulation).

Notes [TODO: add notes]

2.14 fehring2025prefrontal

fehring2025prefrontal (fehring2025prefrontal) — fehring2025prefrontal (fehring2025prefrontal)
literature/fehring2025prefrontal.pdf

- Pseudorandomised cross-over study, $n = 27$ young adults (18–25 y), six consecutive 3-min screen conditions (TV-viewing, social media, gaming) on a phone while seated.
- dlPFC haemodynamics measured continuously with fNIRS (Portalite Mk II); mood states (energy, tension, focus, happiness) rated by VAS before and after each condition.
- HbO increased from baseline most for social media, then gaming, then TV-viewing (all pairwise differences $p < 0.001$).

- HbR increased for gaming but decreased for social media and TV-viewing; HbT increased most for gaming, then social media, and decreased for TV-viewing.
- Focus increased relative to baseline for TV-viewing and gaming but decreased for social media; no significant condition effects on energy, tension or happiness.
- Authors conclude effects are nuanced and context dependent rather than universally beneficial or detrimental, and that fNIRS is feasible for measuring them.
- Takeaway for us: a short-block, within-subject cross-over with 3-min conditions is a workable design template for contrasting auditory environments, and the HbO/HbR/HbT triplet plus a brief subjective scale is a reasonable minimal outcome set.

Notes [TODO: add notes]

2.15 dryden2017speechinnoise

dryden2017speechinnoise (dryden2017speechinnoise) — dryden2017speechinnoise (dryden2017speechinnoise) literature/dryden2017speechinnoise.pdf

- [NOTE: PDF not yet in literature/ — automated download from SAGE and PMC was blocked. Fetch manually from <https://doi.org/10.1177/2331216517744675> and save as literature/dryden2017speechinnoise.pdf.]
- Systematic review and meta-analysis of the association between cognitive performance and speech-in-noise perception in adult listeners.
- Directly relevant to the review question: it establishes that understanding speech in a noisy environment is partly a cognitive, not purely audiological, achievement — motivating the measurement of mental workload rather than performance alone.
- [TODO: Read and extract effect sizes / which cognitive domains (working memory, attention, processing speed) carry the association; then add \fact lines under §3, “Auditory environment and task performance”.]

Notes [TODO: add notes]

3 Facts and statements

[TODO: Bank of citable, manuscript-ready sentences. One \fact{bibkey}{sentence} per line; group by theme. Copy the sentence + the grey \parencite snippet straight into the manuscript.]

3.1 Rationale for mobile neuroimaging (vs. fMRI)

- The acoustic noise generated by the MRI scanner itself alters brain activation during working-memory tasks, confounding fMRI studies of auditory environment and cognition. (tomasi2005fmrinoise) \parencite{tomasi2005fmrinoise}

3.2 Noise, physiology and mental workload

- Acoustic noise is defined as disturbing, unpleasant or unwanted sound, and chronic noise exposure is associated with negative health effects. (**fan2022noise**)
`\parencite{fan2022noise}`
- Physiological responses to noise become significant above roughly 70 dB; 35–65 dB induces annoyance and can disrupt sleep, 66–85 dB can cause psychological, neurological and auditory impairment, and 85–115 dB can cause significant hearing damage. (**fan2022noise**)
`\parencite{fan2022noise}`
- Under noisy conditions participants exert additional effort to maintain task performance; noise intensity increases heart-rate-variability-based stress indices, and higher mental workload decreases EEG alpha power while increasing theta power. (**fan2022noise**)
`\parencite{fan2022noise}`

3.3 Auditory environment and task performance

- [\[TODO: add facts from \(hiwa2018fnirs; gabbard2017fnirs; yun2025anc; pieper2021noise\) as you read them.\]](#)
- [\[TODO: add facts from \(dryden2017speechinnoise\) once the PDF is obtained.\]](#)

3.4 fNIRS as a real-time cognitive-load measure in applied settings

- Subjective cognitive-load instruments are valuable but retrospective and susceptible to unrelated influences, motivating multimodal, real-time objective assessment. (**cannon2026surgery**)
`\parencite{cannon2026surgery}`
- Prefrontal fNIRS metrics—total peak oxygenation, duration-dependent oxygenation, average oxygenation per second and total peak haemoglobin—significantly predicted which of three percutaneous spine surgical workflows a surgeon was performing, with the robot-assisted workflow imposing significantly lower cognitive load. (**cannon2026surgery**)
`\parencite{cannon2026surgery}`
- Even three minutes of screen exposure produces measurable, content-dependent changes in dorsolateral prefrontal HbO, HbR and HbT, demonstrating that fNIRS is sensitive to short, naturalistic environmental manipulations. (**fehring2025prefrontal**)
`\parencite{fehring2025prefrontal}`
- Changes in prefrontal haemodynamics and changes in self-reported focus dissociate across conditions, so haemodynamic load and subjective state should be measured together rather than treated as interchangeable. (**fehring2025prefrontal**)
`\parencite{fehring2025prefrontal}`

3.5 Neural efficiency

- The neural efficiency hypothesis that more able individuals show lower task-related brain activation holds primarily for tasks of low to moderate difficulty; in highly complex tasks, or in well-trained individuals, the relationship can invert such that increased activation accompanies better performance. (**neubauer2009neuralefficiency**)
`\parencite{neubauer2009neuralefficiency}`
- Prefrontal fNIRS signals correlate strongly with perceived mental workload but not with actual task performance, consistent with neural efficiency: individuals differ in how much activation they need to achieve the same outcome. (**causse2017workload**)
`\parencite{causse2017workload}`

- A neural efficiency index may be comparable across individuals performing the same cognitive task, but it does not transfer across different cognitive domains. (cause2017workload)
\parencite{cause2017workload}
- Increasing task difficulty produces increased prefrontal HbO and decreased HbR, supporting fNIRS as a measure of mental effort in both laboratory and ecological settings; however, predicting task performance from fNIRS alone is not feasible because of inter-individual variability in neural efficiency, task difficulty and personality. (cause2017workload)
\parencite{cause2017workload}

[NOTE: Scoping-review methodology refs (Arksey & O'Malley 2005; Levac et al. 2010; Tricco et al. 2018) are in references.bib and forced into the list via \nocite. Still to add: fNIRS-BCI review; neurofeedback review; sonification of neural signals; HAROLD / ageing lateralization; MCI & neurodegeneration monitoring.]